

AI Resume Screening System

An AI-powered Resume Screening System that analyzes resumes against Job Descriptions and ranks candidates based on skill matching and relevance.

Features

- Upload multiple resumes
 - Upload Job Description
 - AI-based candidate analysis
 - Candidate ranking system
 - Match score generation
 - Missing skills detection
 - Resume preview
 - CSV/Excel export support
 - FastAPI backend
 - React/Vite frontend
-

Tech Stack

Backend

- FastAPI
- Python
- SQLAlchemy
- LangChain
- Groq API

Frontend

- React
 - Vite
 - Tailwind CSS
-

Project Structure

```
resume-screening-app/  
├──  
├── backend/  
│   ├── app/  
│   ├── uploads/  
│   └── venv/  
└──
```

```
|  
├── frontend/  
│   ├── src/  
│   └── public/
```

Backend Setup

1. Navigate to backend folder

```
cd backend
```

2. Create Virtual Environment

Linux/macOS

```
python3 -m venv venv  
source venv/bin/activate
```

Windows

```
python -m venv venv  
venv\Scripts\activate
```

3. Generate `requirements.txt` (For Developers)

If `requirements.txt` is not available, install required libraries manually and generate it using:

```
pip freeze > requirements.txt
```

Then install dependencies using:

```
pip install -r requirements.txt
```

4. Create `.env` File

Create a `.env` file inside the `backend` folder.

Example:

```
GROQ_API_KEY=your_groq_api_key  
DATABASE_URL=your_database_connection_url
```

GROQ API Key

Create your API key from:

<https://console.groq.com/keys>

Example Database URLs

SQLite

```
DATABASE_URL=sqlite:///./resume_screening.db
```

PostgreSQL

```
DATABASE_URL=postgresql://username:password@localhost/dbname
```

5. Run Backend Server

```
uvicorn app.main:app --reload
```

Backend server will run at:

```
http://127.0.0.1:8000
```

6. FastAPI Swagger Documentation

Open:

```
http://127.0.0.1:8000/docs
```

to test APIs directly from the browser.

Frontend Setup

1. Navigate to frontend

```
cd frontend
```

2. Install dependencies

```
npm install
```

3. Run frontend

```
npm run dev
```

Frontend runs on:

```
http://localhost:5173
```

API Documentation

FastAPI Swagger Docs:

```
http://127.0.0.1:8000/docs
```

Requirements

- Python 3.10+
- Node.js 18+
- npm
- Git

Author

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